

Ahmed Mohamed

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SUMMARY

Junior Software Engineer with hands-on experience in Full-Stack and Backend Development, Networking, Cloud Computing, and AI Automation. Experienced with React.js, Node.js, Express.js, REST APIs, Prisma, MongoDB, MySQL, JWT, Linux, Git, and n8n. Strong foundation in networking through CCNA training and Cisco Packet Tracer labs. Currently developing expertise in AWS, Docker, DevOps, System Design, Cloud Security, and AI Automation.

EDUCATION

B.Sc. Software Engineering – Capital University

2024 – May 2028

TECHNICAL SKILLS

Languages: C, C++, Java, PHP, JavaScript, SQL

Frontend: React.js, React Router, Tailwind CSS, Framer Motion

Backend: Node.js, Express.js, REST APIs, JWT, Middleware, Authentication

Database: MongoDB, MySQL, Prisma ORM

Cloud/DevOps: AWS, Linux, Docker, Nginx, Reverse Proxy, CI/CD Fundamentals

Networking: TCP/IP, OSI, IPv4/IPv6, Subnetting, VLANs, Routing, Switching, DHCP, DNS, NAT, EtherChannel, LACP, HSRP, Cisco IOS

Concepts: OOP, Data Structures, Operating Systems, Database Systems, REST APIs

Tools: Git, GitHub, Postman, VS Code, Cisco Packet Tracer, Cisco Gns3

PROJECTS

CPU Scheduling Simulator

Java · OOP · GUI

- Developed a desktop application to simulate and compare OS scheduling algorithms (SRTF and Priority Scheduling).
- Implemented scheduling logic handling process arrival times, burst times, and priorities to accurately simulate OS behavior.
- Built a graphical interface to visualize process execution and calculated waiting, turnaround, and completion times.
- Structured the codebase using OOP — separate classes for processes, schedulers, and simulation logic.

ArtScope – Online Art Auction Platform

Laravel · React · PHP · MySQL · JavaScript

- Built a web application where users can register, list artwork, place bids, and track active auctions.
- Developed and consumed REST APIs for communication between the frontend and backend, tested extensively using Postman.
- Implemented backend logic with Laravel (routing, authentication, DB interactions) and a responsive React frontend.
- Designed a relational database schema covering user accounts, artwork listings, auction records, and bid history.
- Designed test cases, identified defects, and verified fixes throughout the development lifecycle.

Full-Stack E-Commerce Application

- Built a full-stack e-commerce application with REST APIs for authentication, products, cart, and orders.
- Implemented JWT authentication, password hashing, and database integration.
- Developed a responsive React.js interface using Tailwind CSS.

Linux & Network Infrastructure Practice

- Configured Node.js applications on Linux and used Nginx as a reverse proxy.
- Practiced VLANs, routing, DHCP, IPv4/IPv6, subnetting, and network troubleshooting.

TRAINING

IEEE Machine Learning Training Program

36 Hours

- Studied data preprocessing, feature selection, model training, and performance evaluation.
- Gained introductory exposure to supervised learning algorithms and the ML development lifecycle.

DEPI — AI Automation & n8n (Digital Egypt Pioneers Initiative) AI Workflows & Automation

- Developed practical knowledge of AI automation and workflow automation using n8n.
- Built automated workflows integrating APIs, AI services, and external applications.

CCNA Training

- Hands-on experience with routing, switching, VLANs, subnetting, IPv4/IPv6, DHCP, DNS, and network troubleshooting.
- Configured EtherChannel/LACP and HSRP using Cisco Packet Tracer.

NTI Summer Program — Route Academy (National Telecommunication Institute)

- Developed skills in communication, teamwork, leadership, problem-solving, negotiation, business writing, and interview preparation.

LANGUAGES

Arabic: Native

English: B2 – Upper Intermediate